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Time is Encoded in the Weights of Finetuned Language Models

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Abstract

We present *time vectors*, a simple tool to customize language models to new time periods. Time vectors are created by finetuning a language model on data from a single time (e.g., a year or month), and then subtracting the weights of the original pretrained model. This vector specifies a direction in weight space that, as our experiments show, improves performance on text from that time period. Time vectors specialized to adjacent time periods appear to be positioned closer together in a manifold. Using this structure, we interpolate between time vectors to induce new models that perform better on intervening and future time periods, without any additional training. We demonstrate the consistency of our findings across different tasks, domains, model sizes, and time scales. Our results suggest that time is encoded in the weight space of finetuned models.

1 Introduction

Temporal variation is a fundamental characteristic of language. As we show in §3, it manifests in language model development as *temporal misalignment*, where deviations in train and test data lead to large performance degradation across different time periods (Luu et al., 2022; Lazaridou et al., 2021; Jaidka et al., 2018, *inter alia*). This necessitates adaptation techniques for customizing models to specific time periods as needed. Designing such techniques is difficult, however, due to the multitude of time scales and the possibility that data from a target time period might be unavailable.

Recent work has shown that the behavior of neural networks can be edited through closed-form interpolation between parameters of finetuned models (Ilharco et al., 2023; Ortiz-Jiménez et al., 2023; Li et al., 2022; Wortsman et al., 2021, *inter alia*). In this work, we demonstrate that weight-space interpolation can also be used to cheaply edit language model behavior over *time*. To this end, we introduce *time vectors* (§4), an extension of task

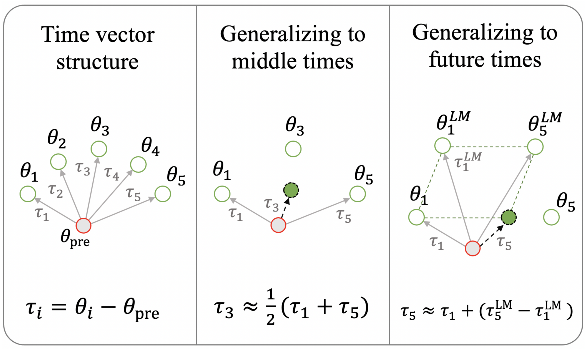

Figure 1: **We present *time vectors*, a simple tool to customize language models to new time periods.** Time vectors (τ_i) specify a direction in weight space that improves performance on text from a time period i. They are computed by subtracting the pretrained weights (θ_{pre}; left panel) from those finetuned to a target time period (θ_i). We can customize model behavior to new time periods (e.g., intervening months or years) by interpolating between time vectors and adding the result to the pretrained model (middle panel). We can also generalize to a future time period j with analogy arithmetic (right panel). This involves combining a task-specific time vector with analogous time vectors derived from finetuned language models (τ_j^{LM}).

vectors (Ilharco et al., 2023). We finetune a pretrained language model on text from a single time period, and then subtract the pretrained weights. This vector represents a direction of movement in weight space that improves performance on text from the target time period.

We analyze the structure of time vectors with temporally organized datasets for language modeling, classification, and summarization (§2). Our results consistently suggest that time vectors are intuitively organized on a manifold; years or months that are closer together in time yield time vectors that are also closer together in weight space. Similarly, we show that temporal degradation in yearly and monthly settings is strongly correlated with the angles between time vectors (§4.2).

We use this structure of time vectors to induce

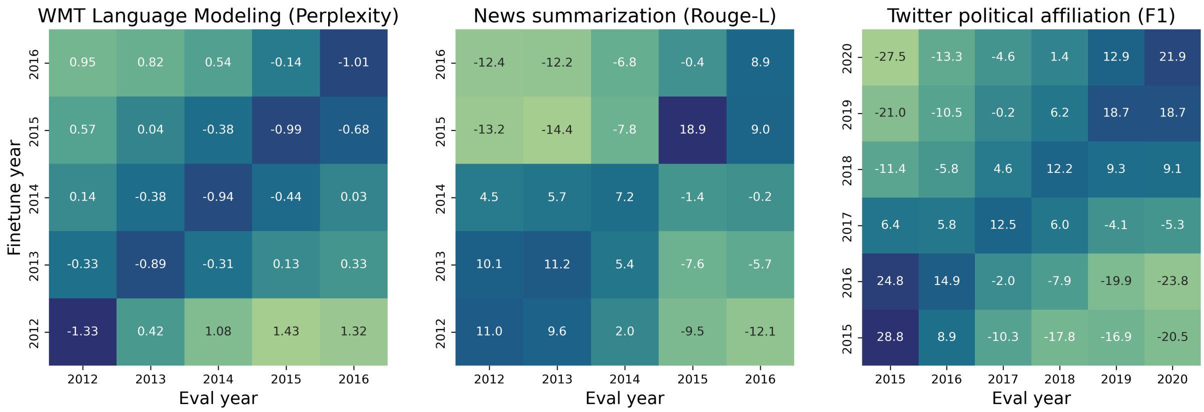

Figure 2: **Model performance degrades linearly year-to-year.** We evaluate language model perplexity (WMT), ROUGE-L (news summarization), and macro F1 (political affiliation classification). Each cell indicates the monthly performance of T5-3B finetuned and evaluated on a *single year* from that task. We report the percentage difference from the average performance for each year, and find linear degradation as finetuning and evaluation years become more misaligned regardless of task. We display similar trends for T5-small and medium, as well as for other domains and tasks, in §A.1. We measure the linearity of these degradations in Appendix Table 4.

models that generalize better to data from new time periods. By interpolating between two time vectors, we discover vectors that, when applied to the pretrained model, improve performance on intervening months or years (§4.3). The structure can also be used to generalize task-specific models across time periods with analogous time vectors specialized to unlabeled data (§4.4).

Our results show that temporal variation is to some extent encoded in the weight space of finetuned models, and that weight interpolation can help customize language models to new time periods. We publicly release our code, data, and over 500 models finetuned on specific time periods.[1]

2 Data and Finetuning

In this section, we describe our datasets and finetuning techniques, which serve as the basis for all subsequent experiments. We finetune language models on multiple time-stratified datasets, which we use to analyze temporal misalignment and build time vectors. Then, we explore different ways of interpolating between time vectors to generalize to new times. See §4.3-4.5 for more details on interpolation strategies.

2.1 Datasets

Language Modeling We create two new time-specific language modeling datasets from unlabeled text in news and Twitter domains. For these datasets, we measure *perplexity* of the model on the test set:

- WMT Language Modeling: We randomly sample 67K $\pm$ 5K articles (47M BPE tokens) of training articles and 3K $\pm$ 0.3K test articles (2.3–2.4M tokens of) from each year 2012–2021 in the English subset of the WMT news dataset (Barrault et al., 2021), from 2012–2016. From the same time range, we also sample 7.1M tokens of training articles and 700–720K tokens of test articles from each month. We are missing WMT train and test splits for August 2012 and May 2016.

- Twitter Language Modeling: We randomly sample 2M $\pm$ 105K training tweets (72–78M tokens BPE tokens) and 100K $\pm$ 5.4K test tweets (3.6-3.9M BPE tokens) from each year in the Internet Archive Twitter Stream Grab ,[2] from 2015–2020. We only use this dataset to study the domain-specificity of time vectors in §4.4.

To understand the level of contamination in our datasets, we measure the overlap between yearly train and test splits in both tasks using a Bloom filter.[3] We find that less than two percent and 0.1 percent of examples in the Twitter and WMT LM test sets, respectively, contain contaminated n-grams.

[1] https://github.com/KaiNylund/lm-weights-encode-time

[2] https://archive.org/details/twitterstream

[3] https://github.com/allenai/bff

Downstream Tasks For downstream tasks, we draw from Luu et al. (2022). We measure each model's performance on the test set in *ROUGE-L* for NewsSum and *macro F1* for PoliAff.

- NewsSum: We use Luu et al. (2022) postprocessing of Grusky et al. (2018) news summarization task. To align with out WMT dataset, we do not bin adjacent years together, creating uniformly sized splits for each year from 2012 to 2016.
- PoliAff: We use the Political Affiliation task from Luu et al. (2022), with uniformly sized datasets for each year from 2015 to 2020.

2.2 Finetuning

To compare the same weight space across tasks, we use pretrained T5 (Raffel et al., 2023) checkpoints for all our experiments. We finetune T5-small, T5-large, and T5-3b on each of our time-stratified datsets. For language modeling, we use the "LM adaptation" objective (Lester et al., 2021).

To reduce the computational burden, we finetune T5-large and T5-3B with Low-Rank Adaptation (LoRA; Hu et al., 2021) and default hyperparameters (q and v attention target modules, $r = 8, \alpha = 32$, dropout = 0.1). When creating time vectors, we merge LoRA weights back into the base model before subtracting the pretrained model.

Across all settings, we use a batch size of 2 with 8 gradient accumulation steps. We finetune for a single epoch on LM splits and three epochs on downstream task splits. Our learning rates across all tasks are 8×10^{-4} for T5-small and T5-large, and 2×10^{-4} for T5-3b. We finetuned models concurrently with a single GPU each; we used 8 2080ti, 4 Titan, and 8 A40 GPUs. In experiments for sections §4.4 and §4.5, we ran evaluations in parallel using available Titan, A40, and A100 GPUs.

3 Revealing Temporal Misalignment at Multiple Time Scales

We begin with an analysis of temporal misalignment using the new set of models and tasks that we consider in this work (§2). These findings set the stage for our creation of time vectors in §4.

3.1 Yearly Degradation is Linear

Previous work on temporal misalignment shows that models degrade over time on a yearly basis.

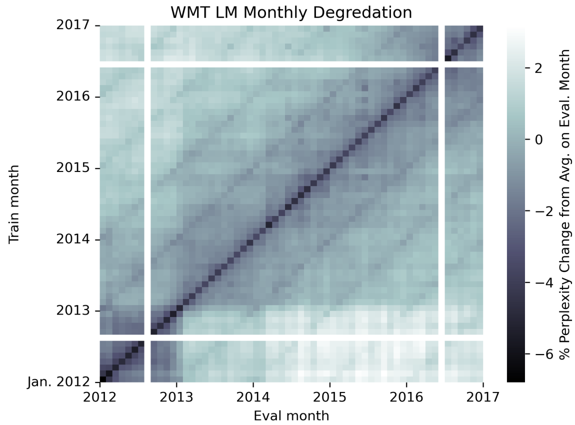

Figure 3: **Monthly temporal degradation has seasonal patterns.** Each cell indicates the monthly performance of T5-small finetuned and evaluated on a *single month* of the WMT dataset. We report the percentage difference in test perplexity from the average on the evaluation month over all finetuned T5-small models (darker is better). The diagonal indicates that each model does best on its finetuning month. Models also do relatively better on the same month in other years, visible as the stripes radiating out from the diagonal every 12 months.

To confirm these results, we finetune T5-small, T5-large, and T5-3b on each yearly split from every dataset. We then evaluate each of these year-finetuned models on every other time split of the test data.

We display heatmaps of temporal misalignment at a yearly scale in Figure 2. We report percent perplexity change from the average on each year to avoid inherent year performance differences. Consistent with past work (Lazaridou et al., 2021; Luu et al., 2022; Longpre et al., 2023), we observe linear patterns of degradation in each task for all model sizes (see Table 4 in the Appendix for more details). Like Luu et al. (2022) show, some tasks, like political affiliation classification, exhibit clearer degradation than others. We quantify these variations in §A.2.

3.2 Monthly Degradation is Seasonal

Next, we turn to month-by-month temporal misalignment, which, to the best of our knowledge, is unexplored. We train T5-small on each WMT LM month split from 2012–2016, resulting in 58 month-finetuned models. We then test every 2012–2016 month model on each month test split for a total of 3,364 evaluations.

As seen in Figure 3, finetuning and evaluating models on specific months in the WMT dataset re-